

Equilibrium Concepts

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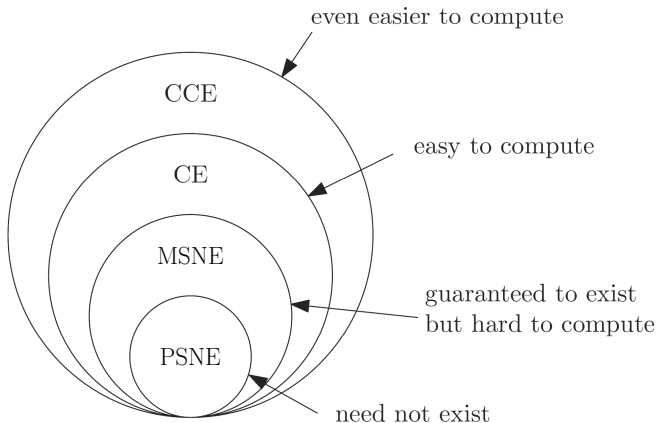


Outline

- 1 Cost Minimization and Payoff Maximization
- 2 Pure Strategy Nash Equilibria (PSNE)
- 3 Mixed Strategy Nash Equilibria (MSNE)
- 4 Correlated Equilibria (CE)
- 5 Coarse Correlated Equilibria (CCE)



A hierarchy of equilibrium concepts



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Cost-Minimization Games

A cost-minimization game has the following ingredients:

- a finite number of k agents;
- a finite set S_i of pure strategies for each agent i ;
- a nonnegative cost function $C_i(\mathbf{s})$ for each agent i .
 - $\mathbf{s} \in S_1 \times S_2 \times \cdots \times S_k$: a **state**, **strategy profile** or **outcome**.

For example, the facility location game, congestion (routing) game, etc.



Payoff-Maximization Games

A **payoff-maximization** game has the following ingredients:

- a finite number of k agents;
- a finite set S_i of pure strategies for each agent i ;
- a nonnegative **payoff** function $\pi_i(\mathbf{s})$ for each agent i .
 - $\mathbf{s} \in S_1 \times S_2 \times \cdots \times S_k$: a **state**, **strategy profile** or **outcome**.

For example, the Rock-Paper-Scissors game, two-party election game, etc.



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Pure (Strategy) Nash Equilibrium (PSNE)

Pure Strategy Nash Equilibrium (PSNE)

A strategy profile $\mathbf{s}$ of a cost-minimization game is a pure strategy Nash equilibrium (PSNE) if for every agent $i \in \{1, 2, \dots, k\}$ and every unilateral deviation $s'_i \in S_i$,

$$C_i(\mathbf{s}) \leq C_i(s'_i, \mathbf{s}_{-i}).$$

- $\mathbf{s}_{-i}$: the vector $\mathbf{s}$ with the i th component removed.
- $\mathbf{s}$ can be viewed as $(s_i, \mathbf{s}_{-i})$ for each $i \in \{1, 2, \dots, k\}$.



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Mixed Strategy Nash Equilibrium (MSNE)

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Distributions $\sigma_1, \dots, \sigma_k$, over strategy sets $S_1, \dots, S_k$ respectively, of a cost-minimization game constitute a mixed Nash equilibrium (MSNE) if for every agent $i \in \{1, 2, \dots, k\}$ and every unilateral deviation $s'_i \in S_i$,

$$\mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s})] \leq \mathbb{E}_{\mathbf{s}_{-i} \sim \sigma_{-i}}[C_i(s'_i, \mathbf{s}_{-i})].$$

- σ : the product distribution $\sigma_1 \times \sigma_2 \times \dots \times \sigma_k$.



Product of Mixed Strategies

Player 1

p_1 rock
 p_2 scissors
 p_3 paper
 probabilities

Player 2

q_1 rock
 q_2 scissors
 q_3 paper
 probabilities

$p_1 q_1$ 0, 0	$p_1 q_2$ 1, -1	$p_1 q_3$ -1, 1
$p_2 q_1$ -1, 1	$p_2 q_2$ 0, 0	$p_2 q_3$ 1, -1
$p_3 q_1$ 1, -1	$p_3 q_2$ -1, 1	$p_3 q_3$ 0, 0

$$\begin{cases} p_1 + p_2 + p_3 = 1. \\ q_1 + q_2 + q_3 = 1. \end{cases}$$

Exercise: Pure Deviations in an MSNE (1/3)

- Consider a finite cost-minimization game $(N, (S_i)_{i \in N}, (C_i)_{i \in N})$, where $N = \{1, 2, \dots, k\}$ and every S_i is finite. Let $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_k)$ be a mixed-strategy profile and $\sigma_{-i} = \prod_{j \neq i} \sigma_j$.
- For every $\rho_i \in \Delta(S_i)$, define

$$\bar{C}_i(\rho_i, \sigma_{-i}) := \mathbb{E}_{\substack{a_i \sim \rho_i \\ s_{-i} \sim \sigma_{-i}}} [C_i(a_i, s_{-i})].$$

- Let δ_{a_i} assign probability one to a_i (i.e., pure strategy).

Prove that, for every $\tau_i \in \Delta(S_i)$,

$$\bar{C}_i(\tau_i, \sigma_{-i}) = \sum_{a_i \in S_i} \tau_i(a_i) \bar{C}_i(\delta_{a_i}, \sigma_{-i}).$$



Exercise: Pure Deviations in an MSNE (2/3)

- Consider a finite cost-minimization game $(N, (S_i)_{i \in N}, (C_i)_{i \in N})$, where $N = \{1, 2, \dots, k\}$ and every S_i is finite. Let $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_k)$ be a mixed-strategy profile and $\sigma_{-i} = \prod_{j \neq i} \sigma_j$.
- For every $\rho_i \in \Delta(S_i)$, define

$$\bar{C}_i(\rho_i, \sigma_{-i}) := \mathbb{E}_{\substack{a_i \sim \rho_i \\ s_{-i} \sim \sigma_{-i}}} [C_i(a_i, s_{-i})].$$

- Let δ_{a_i} assign probability one to a_i (i.e., pure strategy).

Suppose that τ_i is a profitable mixed deviation:

$$\bar{C}_i(\tau_i, \sigma_{-i}) < \bar{C}_i(\sigma_i, \sigma_{-i}).$$

Prove that some $a_i \in S_i$ with $\tau_i(a_i) > 0$ satisfies

$$\bar{C}_i(\delta_{a_i}, \sigma_{-i}) < \bar{C}_i(\sigma_i, \sigma_{-i}).$$



Exercise: Pure Deviations in an MSNE (3/3)

Prove that σ is a MSNE if and only if, for every player i and every $a_i \in S_i$,

$$\mathbb{E}_{s \sim \sigma} [C_i(s)] \leq \mathbb{E}_{s_{-i} \sim \sigma_{-i}} [C_i(a_i, s_{-i})].$$

- In particular, it suffices to check all pure unilateral deviations.

Hint

The expected cost of a mixed deviation is a weighted average of the expected costs of its component pure strategies.



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Correlated Equilibrium (CE)

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A **distribution** σ on the set $S_1 \times S_2 \times \dots \times S_k$ of outcomes of a cost-minimization game is a correlated equilibrium (CE) if

- for every agent $i \in \{1, 2, \dots, k\}$,
- for every recommended action $a_i \in S_i$ with $\Pr_{s \sim \sigma}[s_i = a_i] > 0$, and
- for every unilateral deviation $s_i = a'_i \in S_i$,

we have

$$\mathbb{E}_{s \sim \sigma}[C_i(s) \mid s_i = a_i] \leq \mathbb{E}_{s \sim \sigma}[C_i(a'_i, s_{-i}) \mid s_i = a_i].$$



Stop or Go?

Matrix of costs

	Stop	Go
Stop	1, 1	1, 0
Go	0, 1	5, 5

- Two PSNE.

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- Two PSNE.

Stop or Go?

Matrix of costs

	Stop	Go
Stop	prob. = 0 1, 1	prob. = 1/2 1, 0
Go	prob. = 1/2 0, 1	prob. = 0 5, 5

- A CE for example.
- Cannot correspond to a MSNE.



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Matrix of costs

	Stop	Go
Stop	prob. = 0 1, 1	prob. = 1/2 1, 0
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- A CE for example.
- Cannot correspond to a MSNE.

Game of Chicken

- A.k.a. Hawk-Dove Game.
 - A model of conflict for two players.

	Dare	Chicken
Dare	0, 0	7, 2
Chicken	2, 7	6, 6

- Two PSNE & One MSNE.
- The expected utility of each player in the MSNE:

$$\frac{1}{3} \cdot 2 + \frac{2}{3} \cdot 6 = \frac{14}{3}$$



Game of Chicken

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$$\frac{1}{3} \cdot \frac{2}{3} \cdot 7 + \frac{2}{3} \cdot \frac{1}{3} \cdot 2 + \frac{2}{3} \cdot \frac{2}{3} \cdot 6 = \frac{14}{3}.$$



Game of Chicken

- A correlated equilibrium.
 - Check that it is an equilibrium if a player is assigned “Dare”.
 - Check that it is an equilibrium if a player is assigned “Chicken Out”.

	Dare	Chicken
Dare	$\text{prob.} = 0$ 0, 0	$\text{prob.} = 1/3$ 7, 2
Chicken	$\text{prob.} = 1/3$ 2, 7	$\text{prob.} = 1/3$ 6, 6

- The expected utility for each player:
 $7 \cdot (1/3) + 2 \cdot (1/3) + 6 \cdot (1/3) = 5.$

Game of Chicken

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Game of Chicken

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Coarse Correlated Equilibrium (CCE)

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A **distribution σ on the set $S_1 \times S_2 \times \dots \times S_k$ of outcomes** of a cost-minimization game is a coarse correlated equilibrium (CE) if for every agent $i \in \{1, 2, \dots, k\}$ and every unilateral deviation $s'_i \in S_i$,

$$\mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s})] \leq \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s'_i, \mathbf{s}_{-i})].$$

CE $\subseteq$ CCE?

$$\begin{aligned} \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s})] &= \sum_{a \in S_i} \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s}) \mid s_i = a] \Pr[s_i = a] \\ &\leq \sum_{a \in S_i} \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s'_i, \mathbf{s}_{-i}) \mid s_i = a] \Pr[s_i = a] \\ &= \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s'_i, \mathbf{s}_{-i})] \end{aligned}$$



Coarse Correlated Equilibrium (CCE)

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A **distribution σ on the set $S_1 \times S_2 \times \dots \times S_k$ of outcomes** of a cost-minimization game is a coarse correlated equilibrium (CE) if for every agent $i \in \{1, 2, \dots, k\}$ and every unilateral deviation $s'_i \in S_i$,

$$\mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s})] \leq \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s'_i, \mathbf{s}_{-i})].$$

CE $\subseteq$ CCE?

$$\begin{aligned}\mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s})] &= \sum_{a \in S_i} \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s}) \mid s_i = a] \Pr[s_i = a] \\ &\leq \sum_{a \in S_i} \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s'_i, \mathbf{s}_{-i}) \mid s_i = a] \Pr[s_i = a] \\ &= \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s'_i, \mathbf{s}_{-i})]\end{aligned}$$



CCE Example

	A	B	C
A	prob. = $\frac{1}{3}$ 1, 1	-1, -1	0, 0
B	-1, -1	prob. = $\frac{1}{3}$ 1, 1	0, 0
C	0, 0	0, 0	prob. = $\frac{1}{3}$ -1.1, -1.1

- The payoff for each player (playing according to this distribution):
 $\frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 1 - \frac{1}{3} \cdot 1.1 = 0.3.$
- A player playing fixed A or B while the opponent randomized according to this distribution: $\frac{1}{3} \cdot 1 - \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 0 = 0.$
- A player playing fixed C while the opponent randomized according to this distribution: $\frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot (-1.1) < 0.$



CCE Example

	A	B	C
A	prob. = 1/3 1, 1	-1, -1	0, 0
B	-1, -1	prob. = 1/3 1, 1	0, 0
C	0, 0	0, 0	prob. = 1/3 -1.1, -1.1

- The payoff for each player (playing according to this distribution):
 $\frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 1 - \frac{1}{3} \cdot 1.1 = 0.3$.
- A player playing fixed A or B while the opponent randomized according to this distribution: $\frac{1}{3} \cdot 1 - \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 0 = 0$.
- A player playing fixed C while the opponent randomized according to this distribution: $\frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot (-1.1) < 0$.



CCE Example

	A	B	C
A	$\text{prob.} = 1/3$ $1, 1$	$-1, -1$	$0, 0$
B	$-1, -1$	$\text{prob.} = 1/3$ $1, 1$	$0, 0$
C	$0, 0$	$0, 0$	$\text{prob.} = 1/3$ $-1.1, -1.1$

- The payoff for each player (playing according to this distribution):
 $\frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 1 - \frac{1}{3} \cdot 1.1 = 0.3.$
- A player playing fixed A or B while the opponent randomized according to this distribution: $\frac{1}{3} \cdot 1 - \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 0 = 0.$
- A player playing fixed C while the opponent randomized according to this distribution: $\frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot (-1.1) < 0.$



CCE Example

	A	B	C
A	prob. = 1/3 1, 1	-1, -1	0, 0
B	-1, -1	prob. = 1/3 1, 1	0, 0
C	0, 0	0, 0	prob. = 1/3 -1.1, -1.1

- The payoff for each player (playing according to this distribution):
 $\frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 1 - \frac{1}{3} \cdot 1.1 = 0.3.$
- A player playing fixed *A* or *B* while the opponent randomized according to this distribution: $\frac{1}{3} \cdot 1 - \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 0 = 0.$
- A player playing fixed *C* while the opponent randomized according to this distribution: $\frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot (-1.1) < 0.$



CCE Example

	A	B	C
A	prob. = 1/3 1, 1	-1, -1	0, 0
B	-1, -1	prob. = 1/3 1, 1	0, 0
C	0, 0	0, 0	prob. = 1/3 -1.1, -1.1

- A player playing fixed C and the strategy profile follows this distribution:
 $\Rightarrow$ deviation is possible.
 - Not a CE.



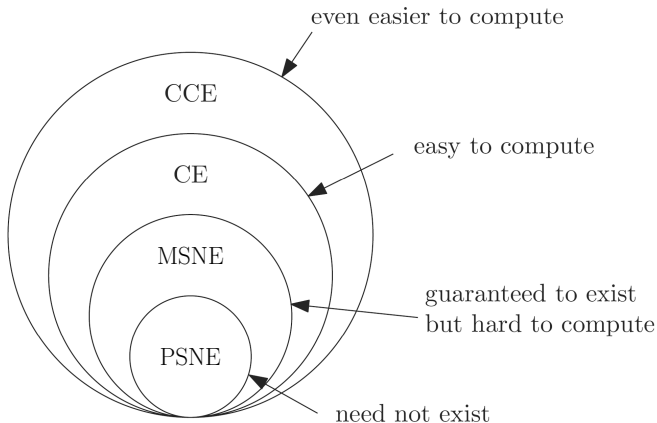
CCE Example

	A	B	C
A	prob. = 1/3 1, 1	-1, -1	0, 0
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Discussions

